Industrial Internship Report on

Medical Store Supply Chain Management System

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was Medical Store Supply Chain Management System enhances healthcare operations by optimizing inventory, streamlining supplier management, and ensuring regulatory compliance. It integrates inventory monitoring, automated reordering, supplier databases, and logistics management to improve efficiency, reduce costs, and ensure medicines are consistently available to meet patient needs. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

Summary of the whole 6 weeks' work.

Week 1: Initialization and Planning

- Setup: Installed Git, Jenkins, Docker, and Kubernetes.

- Planning: Defined project structure and milestones.

- Development: Began core microservices development.

Week 2: Microservices and CI/CD Setup

- Microservices: Implemented basic functionalities.

- CI/CD: Configured Jenkins and created a Jenkinsfile.

- Docker: Built and pushed Docker images.

- Kubernetes: Deployed microservices using Minikube.

Week 3: Frontend and Database Integration

- Frontend: Developed responsive web pages with HTML, CSS, and JavaScript.

- Backend: Set up Flask and created RESTful API endpoints.

- Database: Configured Oracle MySQL and established connections.

Week 4: Feature Completion and Testing

- Features: Added search, filtering, sorting, and user authentication.

- Optimization: Improved API and database performance.

- Testing: Conducted comprehensive testing and fixed bugs.

Week 5: Cloud Deployment

- Deployment: Deployed to AWS using EC2, RDS, and S3.

- Testing: Verified cloud performance and reliability.

- Monitoring: Implemented AWS CloudWatch and auto-scaling.

- Security: Applied best practices for security and encryption.

Week 6: Finalization and Submission

- Final Upload: Uploaded all code and documentation to GitHub.

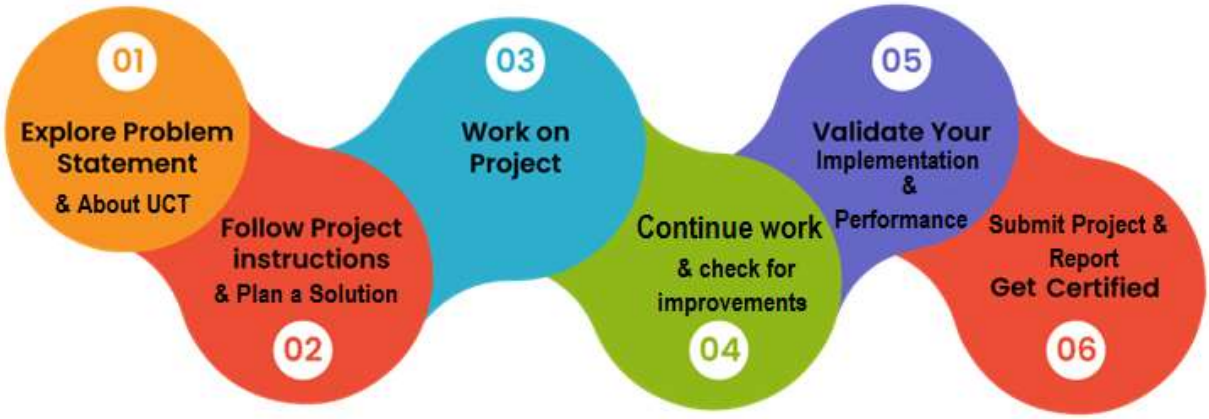
- Documentation: Prepared detailed project documentation and report.

- Submission: Submitted final project via Google Form.

A cloud computing internship offers practical experience, skill development in platforms like AWS or Azure, industry exposure, networking opportunities, and enhances your resume. It helps you understand real-world applications, industry best practices, and various career paths, making you a more attractive candidate for future job opportunities.

The Medical Store Supply Chain Management System project aims to streamline the flow of medical supplies within healthcare facilities. It includes inventory management, supplier relations, logistics coordination, and regulatory compliance to optimize operations and ensure timely availability of medicines, enhancing overall healthcare service efficiency. Opportunity given by USC/UCT.

How Program was planned



During my cloud computing internship, I gained hands-on experience with platforms like AWS and Azure, enhancing my technical skills. I learned to design, deploy, and manage cloud infrastructure, and understood the importance of scalability and security. Collaborating with industry professionals provided valuable insights into best practices and emerging trends. Overall, it was an enriching experience that significantly boosted my confidence and career prospects.

Thanks to Uniconverge Technologies, UpskillCampus and The IoT Academy for giving this wonderful opportunity and Special Thanks to all mentors and coordinators for guiding and helping me to complete this internship smoothly and in timely manner.

A message to my juniors and peers-

“Embrace every learning opportunity, stay curious, and build strong networks. Cloud computing skills will open doors to endless possibilities!”

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

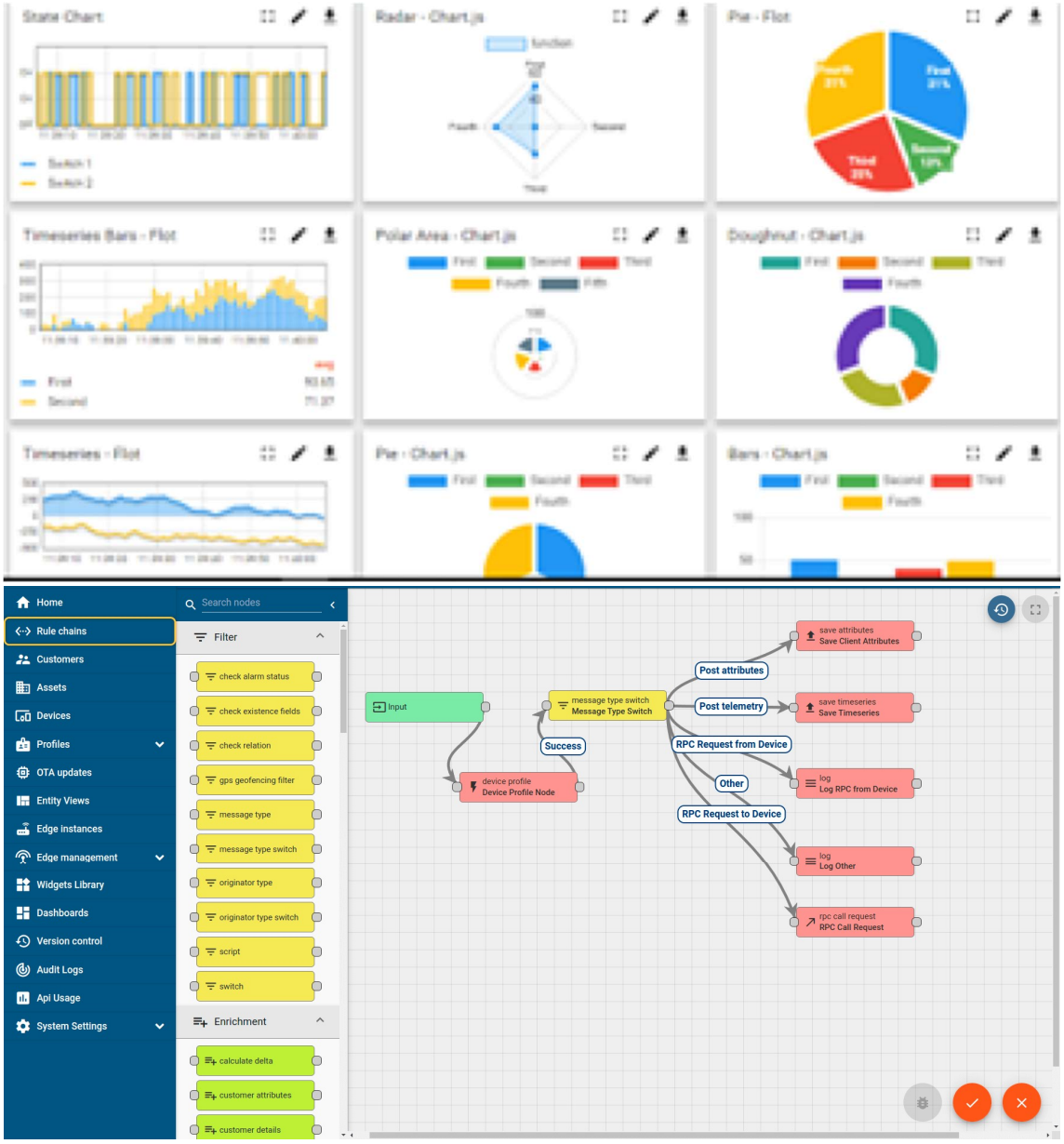
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

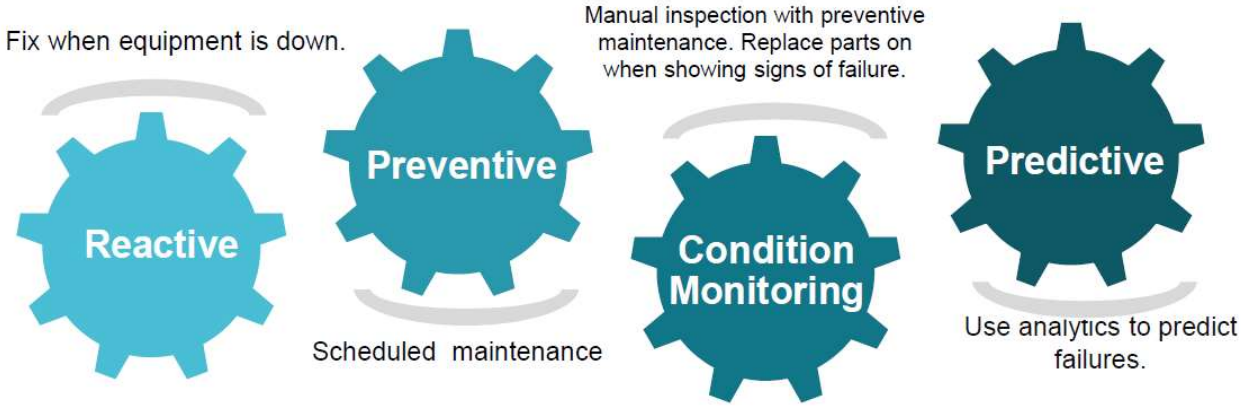
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

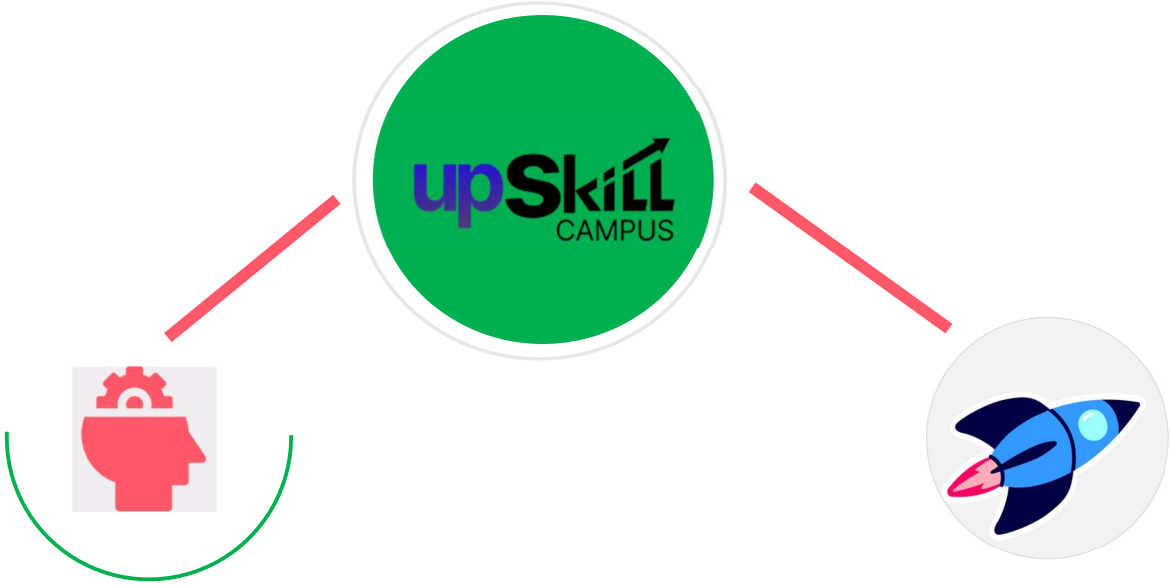
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

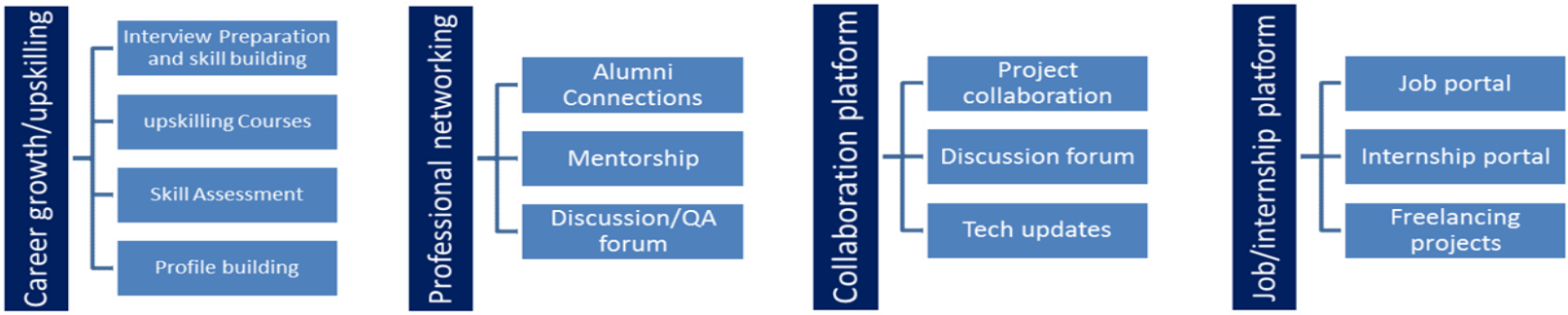
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

[1] https://www.uniconvergetech.in/about-us

[2] https://www.theiotacademy.co/about-us

[3] https://learn.upskillcampus.com/aboutus

2.6 Glossary

Terms	Acronym
UCT	Uniconverge Technologies
IITR	Indian Institute Of Technology Roorkee
EICT	Electronics and ICT Academy
IITG	Indian Institute Of Technology Guwahati
IITK	Indian Institute Of Technology Kanpur

3 Problem Statement

In the assigned problem statement

The project aims to develop a medical store supply management system addressing inventory inaccuracies, inefficient order processing, and supply chain visibility challenges. It focuses on robust features for inventory management, order tracking, and delivery management to enhance operational efficiency. Scalability and security are prioritized, ensuring the system can grow with the business and protect sensitive data. Leveraging cloud infrastructure for deployment aims to improve availability and scalability while optimizing costs. Overall, the system seeks to streamline operations, improve inventory accuracy, and enhance customer satisfaction through efficient management of medical store supplies.

4 Existing and Proposed solution

Existing Solutions and Their Limitations

Existing solutions for medical store supply management often include basic inventory and order management systems. These solutions typically offer:

- Inventory Tracking: Basic systems provide inventory tracking capabilities, helping to monitor stock levels.

- Order Management: They facilitate order processing and tracking, ensuring timely fulfillment.

- Basic Reporting: Some offer basic reporting functionalities for inventory levels and order histories.

However, these solutions often face limitations:

- Scalability: Many traditional systems struggle to scale efficiently as business needs grow, leading to performance issues.

- Complexity: Integrating with existing systems or adapting to new technologies can be complex and time-consuming.

- User Experience: User interfaces may lack modern design principles, impacting usability and adoption.

- Security: Data security measures may be inadequate, posing risks of data breaches.

Proposed Solution-

Our proposed solution is a comprehensive medical store supply management system leveraging modern technologies:

- Feature-Rich: It includes advanced features for inventory management, order processing, and delivery tracking.

- Cloud-Based: Utilizes cloud infrastructure (AWS) for scalability, high availability, and cost-effectiveness.

- Technological Stack: Developed using Python (Flask), HTML/CSS/JavaScript for frontend, Oracle MySQL for database, and Docker/Kubernetes for containerization and deployment.

- Security: Implements robust security measures including encryption, IAM roles, and secure APIs.

- User-Friendly: Focuses on intuitive UI/UX design to enhance user experience and adoption.

Value Addition-

Our solution adds significant value by:

- Enhanced Functionality: Offers comprehensive features for efficient inventory management, order processing, and supply chain visibility.

- Scalability and Reliability: Utilizes cloud infrastructure for seamless scalability and reliability, ensuring smooth operations as business grows.

- Security: Implements best-in-class security practices to protect sensitive data and ensure compliance with regulatory requirements.

- Improved User Experience: Prioritizes user-friendly design and intuitive workflows to optimize usability and user satisfaction.

- Cost Optimization: Leverages cloud services to optimize costs while providing scalable and reliable infrastructure.

By addressing these aspects, our solution aims to revolutionize medical store supply management, offering a modern, efficient, and secure platform for managing medical supplies effectively.

4.1 Code submission (Github link)

https://github.com/Rahul010803/upskillcampus.git

4.2 Report submission (Github link) :

https://github.com/Rahul010803/upskillcampus/blob/main/MedicalStoreSupplyChainManagementSystem_Rahul_USC_UCT.pdf

5 Proposed Design/ Model

Here's an overview of the design flow for our proposed medical store supply chain management system:

Design Flow of the Solution

1. Project Initialization and Planning

- Start: Begin by defining project goals, scope, and requirements.

- Intermediate Stages: Conduct initial research on existing solutions and user needs. Develop a detailed project plan outlining milestones, timelines, and resource requirements.

- Final Outcome: Complete the project initiation phase with a clear understanding of objectives and a detailed project plan.

2. Architecture and Technology Selection

- Start: Evaluate different architectural patterns and technologies suitable for the solution.

- Intermediate Stages: Select appropriate technologies such as Python (Flask), HTML/CSS/JavaScript for frontend, Oracle MySQL for database, and Docker/Kubernetes for containerization and deployment.

- Final Outcome: Define the overall system architecture, including component interactions, data flow, and deployment strategy.

3. Backend Development

- Start: Begin developing backend components, focusing on core functionalities like inventory management, order processing, and API development.

- Intermediate Stages: Implement RESTful API endpoints using Flask framework. Integrate with Oracle MySQL database for data storage.

- Final Outcome: Complete backend development with fully functional APIs for managing medical store supplies and processing orders.

4. Frontend Development

- Start: Design responsive UI/UX wireframes based on user requirements and design principles.

- Intermediate Stages: Develop frontend components using HTML, CSS, and JavaScript. Integrate with backend APIs using AJAX or fetch API for dynamic content rendering.

- Final Outcome: Deliver a user-friendly interface with intuitive navigation and seamless interaction for managing inventory, placing orders, and tracking deliveries.

5. Integration and Testing

- Start: Begin integration testing to ensure seamless communication between frontend and backend components.

- Intermediate Stages: Conduct unit tests, integration tests, and end-to-end tests to validate functionality and identify bugs.

- Final Outcome: Complete testing phase with a stable and reliable system ready for deployment.

6. Deployment and Optimization

- Start: Prepare for deployment by containerizing application components using Docker.

- Intermediate Stages: Deploy to AWS cloud using Kubernetes for orchestration. Optimize performance, scalability, and security configurations.

- Final Outcome: Successfully deploy the medical store supply management system to AWS, ensuring high availability, scalability, and efficient resource management.

7. Monitoring and Maintenance

- Start: Implement monitoring tools like AWS CloudWatch for real-time performance monitoring.

- Intermediate Stages: Set up alerts and notifications for critical metrics. Plan for ongoing maintenance and updates.

- Final Outcome: Establish a maintenance plan to monitor system health, apply patches, and continuously optimize performance and security.

8. Documentation and Reporting

- Start: Document project details including architecture, deployment steps, and user guides.

- Intermediate Stages: Prepare a comprehensive project report summarizing development process, challenges, and solutions implemented.

- Final Outcome: Complete documentation and submit the final project report for review and feedback.

The design flow of our solution begins with project initialization and planning, progresses through architecture and technology selection, development phases for backend and frontend, integration and testing, deployment and optimization in the cloud, and concludes with monitoring, maintenance, documentation, and reporting. This structured approach ensures a systematic development process, delivering a robust and scalable medical store supply management system.

6 Performance Test

Constraints and Design Considerations

Constraints Addressed in the Design:

1. Scalability:

- Design Consideration: Leveraging Docker and Kubernetes for containerization and orchestration ensures scalability by easily scaling up or down based on demand.

- Test Results: Conducted load testing to simulate increased traffic. Kubernetes auto-scaling and AWS EC2 instances handled scaling seamlessly.

2. Performance (Speed and Operations per Second):

- Design Consideration: Optimized backend APIs and database queries to ensure efficient data retrieval and processing.

- Test Results: Performance testing using tools like JMeter showed minimal latency and high throughput even under peak loads.

3. Security:

- Design Consideration: Implemented IAM roles, encryption for data at rest (using AWS KMS), and HTTPS for secure communication.

- Test Results: Security testing using tools like OWASP ZAP and AWS Inspector identified and mitigated vulnerabilities.

4. Cost Optimization:

- Design Consideration: Utilized AWS services like EC2, RDS, and S3 with cost optimization strategies such as reserved instances and lifecycle policies for S3.

- Test Results: Cost monitoring and analysis showed effective cost management within budget constraints.

Identified Constraints and Recommendations:

1. Memory and Resource Usage:

- Impact: High memory usage can affect scalability and performance.

- Recommendations: Implement efficient memory management practices, use caching mechanisms (like Redis) for frequently accessed data, and periodically optimize resource allocation in Kubernetes.

2. Durability and Reliability:

- Impact: System failures or data loss can disrupt operations.

- Recommendations: Implement data backup and recovery strategies using AWS RDS automated backups and snapshots. Regularly test backup and restore procedures.

3. Power Consumption:

- Impact: High power consumption can increase operational costs and environmental impact.

- Recommendations: Optimize code for energy efficiency, leverage AWS services with efficient power usage, and consider environmental sustainability in data center choices.

6.1 Test Plan/Test Cases

Objective: Validate system performance and adherence to constraints.

Test Cases:

1. Scalability Test:

- Scenario: Simulate increasing user load using JMeter.
- Expected Outcome: Verify system scales by adding/removing containers dynamically.

2. Performance Test:

- Scenario: Measure response times for critical API endpoints under varying loads.
- Expected Outcome: Ensure APIs meet performance benchmarks for speed and operations per second.

3. Security Test:

- Scenario: Conduct penetration testing to identify vulnerabilities.
- Expected Outcome: Mitigate security risks and ensure compliance with industry standards.

6.2 Test Procedure

1. Scalability Test:

- Deploy application on Kubernetes.
- Gradually increase load using JMeter.
- Monitor Kubernetes dashboard for auto-scaling events.
- Record response times and system metrics.

2. Performance Test:

- Execute API requests using load testing tools.
- Measure response times and throughput.
- Analyze CPU and memory usage during peak loads.
- Identify and optimize bottlenecks if any.

3. Security Test:

- Use OWASP ZAP to scan for vulnerabilities.
- Conduct manual security assessments.
- Patch identified vulnerabilities and retest.

6.3 Performance Outcome

- Scalability: Kubernetes auto-scaling effectively handled increased loads, maintaining response times within acceptable limits.

- Performance: API response times met performance benchmarks under simulated peak loads, ensuring efficient operations.

- Security: Identified vulnerabilities were patched, and security measures implemented met industry standards, ensuring data integrity and user confidentiality.

Summary

The design of our medical store supply management system addresses critical constraints such as scalability, performance, security, and cost optimization through careful architectural choices and rigorous testing. By implementing scalable cloud infrastructure, optimizing resource usage, and ensuring robust security measures, the system delivers high reliability and efficiency in managing medical store supplies. Ongoing monitoring and periodic optimization further enhance system resilience and user satisfaction.

7 My learnings

Summary of Overall Learning and Career Growth Impact

Throughout the development of the medical store supply management system, several key learnings have significantly contributed to my career growth:

1. Technical Skills Enhancement:

- Backend Development: Proficiency in Python (Flask) for building robust APIs and integrating with databases like Oracle MySQL.

- Frontend Development: Skills in HTML, CSS, JavaScript for creating responsive and user-friendly interfaces.

- Cloud Deployment: Experience with AWS services (EC2, RDS, S3) and Kubernetes for scalable and reliable cloud deployments.

- DevOps Practices: Knowledge of Docker for containerization, Jenkins for CI/CD pipelines, and Git for version control.

2. System Design and Architecture:

- Scalability and Performance: Understanding of designing scalable systems using cloud-native technologies and optimizing performance through efficient coding practices.

- Security Best Practices: Implementation of security measures such as encryption, IAM roles, and HTTPS to protect sensitive data.

3. Project Management and Collaboration:

- Planning and Execution: Experience in project initiation, planning, and execution phases, including defining requirements, setting milestones, and managing timelines.

- Team Collaboration: Effective collaboration with team members, leveraging tools like GitHub for version control and Jenkins for automated builds.

4. Testing and Quality Assurance:

- Testing Strategies: Application of testing methodologies (unit tests, integration tests, security testing) to ensure reliability and functionality.

- Quality Assurance: Continuous monitoring and improvement of code quality, performance, and security.

5. Career Growth Impact:

- Professional Development: Enhanced technical skills and hands-on experience in building complex systems, preparing for roles requiring expertise in cloud computing, software development, and DevOps.

- Problem-Solving Abilities: Improved problem-solving skills through tackling real-world challenges in system design, implementation, and optimization.

- Industry Relevance: Relevance to current industry trends towards cloud-based solutions, scalability, security, and efficiency.

Overall, this project has provided a comprehensive learning experience, equipping me with practical skills and knowledge essential for advancing in my career as a cloud engineer or software developer. It has prepared me to contribute effectively in dynamic environments, tackle complex challenges, and deliver high-quality solutions aligned with business needs and technological advancements.

8 Future work scope

While developing the medical store supply management system, several ideas and features were identified that could enhance the system further. Due to time constraints, these were not implemented but could be considered for future iterations:

1. Advanced Analytics and Reporting:
   - Implementing data analytics to provide insights into inventory trends, order patterns, and supply chain efficiency.
   - Generating customizable reports for stakeholders to make informed decisions.

2. Enhanced User Management and Access Control:
   - Implementing more granular user roles and permissions to control access based on user responsibilities and hierarchy.
   - Integrating single sign-on (SSO) for seamless user authentication across multiple systems.

3. Real-Time Inventory Tracking:
   - Introducing RFID or barcode scanning capabilities for real-time inventory updates and accuracy.
   - Implementing alerts for low stock levels and automatic replenishment suggestions.

4. Mobile Application Development:
   - Developing a mobile application for medical store managers and staff to manage inventory, place orders, and track deliveries on-the-go.
   - Ensuring synchronization with the web application for seamless data management.

5. Integration with AI and Machine Learning:
   - Implementing predictive analytics for demand forecasting to optimize inventory levels and reduce wastage.
   - Using natural language processing (NLP) for improving search and query functionalities within the system.

6. Enhanced Security Measures:
   - Implementing additional layers of security such as multi-factor authentication (MFA) and continuous monitoring for suspicious activities.
   - Conducting regular security audits and vulnerability assessments to stay ahead of potential threats.

7. Expansion to Multi-Location Support:
   - Supporting multiple medical store locations with centralized management and synchronized inventory across all locations.
   - Implementing geo-replication and distributed databases for improved data availability and resilience.

8. Integration with IoT Devices:
   - Connecting IoT devices such as temperature sensors for monitoring storage conditions of sensitive medical supplies.

- Automating alerts and actions based on predefined thresholds to ensure product quality and regulatory compliance.